

RAKSHITH KUMAR K.N

Senior AI / ML Engineer | Generative AI | Production ML Systems | LLMops | Multi-Cloud AI | Team Leadership
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PROFESSIONAL SUMMARY

Senior AI/ML Engineer with **5+ years** delivering **\$2.3M+ measurable business impact** across enterprise security, healthcare, and financial services. Built production MCP servers connecting LLM agents to live enterprise tools (HappyFox, Jira) for real-time context grounding, and contributed to the MCP open-source ecosystem. Alignment engineering practitioner with end-to-end **RLHF experience** — from reward model design and **DPO preference optimization** on 72B-parameter medical LLMs to custom **LLM-as-judge** evaluation frameworks deployed in Fortune 500 security and financial services pipelines. Proven team leader who mentored **5+ junior engineers**, led cross-functional squads of 4-8 engineers across ML, backend, and frontend. Pioneered **LLM-supervised Knowledge Distillation** (GPT-4 Teacher to student ensemble) compressing 4+ hour firewall validation to 2-5 seconds. Multi-cloud architect across **AWS, Azure, and GCP** with **\$50K+/month** in cloud savings. Deep expertise in **multi-agent orchestration**, **GraphRAG**, **LLMOps**, and **regulated-domain AI** (HIPAA, SOC 2). **Zero PII leakage** across 50K+ financial documents.

WORK EXPERIENCE

Senior AI / ML Engineer

Oct 2024 - Present

Gruve AI, Bengaluru

- **Custom Eval Framework + Knowledge Distillation (Cisco ASA):** Architected **GPT-4-supervised evaluation and distillation pipeline** — designed custom eval framework using GPT-4 as judge to score student outputs via KL-divergence loss with **confidence-threshold gating** (per-field precision targets); trained 3 student models achieving **zero LLM calls at inference**. Compressed **4+ hour firewall validation to 2-5 sec** (98% reduction), enabling **Fortune 500 migration** of 200+ firewalls 10x faster. Led 4 engineers. [PyTorch, BERT, GPT-4, LangGraph, Neo4j, Qdrant, AWS Bedrock]
- **Production MCP Server Development:** Built production **MCP servers** connecting LLM agents to HappyFox, Jira, and Confluence for real-time enterprise tool integration, eliminating **85% of manual data fetch latency**. Led 3-person squad; adopted across multiple internal teams. Applied DSPy for prompt optimization. [MCP SDK, FastAPI, TypeScript, Go, WebSocket, DSPy]
- **IDP Supply Chain Agent (Scania):** Achieved **p95 <300ms** on hybrid NL-to-SQL + semantic search across shipping documents by building **5-stage confidence-gated agent harness** (Classifier, Extraction, Auto-Accept, Router, SQL Agent) with **Claude via AWS Bedrock** and MinerU VLM extraction fallback; indexed in Elasticsearch. Designed **custom confidence-gated eval framework** — per-field precision thresholds act as automated **LLM-as-judge** layer, routing low-confidence extractions to HITL queue; reduced human review workload by **73%** with zero critical-field errors. [AWS Bedrock, Claude, FastAPI, Elasticsearch, MinerU VLM, PyMuPDF, SQLAlchemy 2, React, TypeScript, Docker]
- **TechGenie Voice Agent (MCP-Powered):** Achieved **42% reduction in IT resolution time** (500+ weekly interactions) by building **MCP-connected LLM agents** with NeMo Guardrails for safe output control and DSPy optimization reducing hallucination by 34%. Led 4-engineer team. [FastAPI, Edge TTS, MongoDB, MCP, LangGraph, NeMo Guardrails, DSPy]
- **Enterprise AI Search (GSC):** Served **10K+ queries/day** at **p95 <300ms** with **99.9% uptime** by building hybrid **graph + vector retrieval** (1M+ graph nodes, 50M embeddings). Mentored 2 engineers. [Neo4j, Qdrant, AWS Bedrock, ECS, LangSmith, Prometheus]
- **AI Safety - PII Redaction (J.P. Morgan):** Achieved **zero PII leakage** across 50K+ financial documents (99.2% precision, **SOC 2 passed**) by architecting NLP redaction pipeline; scaled to 2,000+ docs/day. [spaCy, Presidio, GLINER, AWS, Azure]
- **GUIS Meeting Intelligence Platform:** Led **5-engineer team** to build 6-service platform for real-time meeting insights across **Google Meet** and **MS Teams**, reducing cross-team meeting follow-up time by **45% across 200+ weekly meetings**. [Node.js, React, FastAPI, Docker, Gemini, Deepgram, GCP, Whisper]
- **Multi-Cloud FinOps:** Saved **\$18K/month** (62% Azure compute reduction) by migrating VMs to event-driven containers across **AWS (ECS, Bedrock, SageMaker)**, **Azure (Container Apps)**, and **GCP (Vertex AI)**. 99.5% SLA. [Terraform, Docker, Kubernetes, MLflow, Prometheus, LangSmith, GitHub Actions, CI/CD]
- **Team Leadership and Mentoring:** Mentored **5 junior ML engineers** (2 promoted within 8 months); established org-wide documentation standards, technical design reviews, and onboarding curriculum.

AI Engineer

Aug 2023 - Oct 2024

August AI, Bengaluru

- **Multi-Agent Patient Interview System:** Raised diagnostic accuracy **67% to 91%** (5,000+ validated sessions, 2,500+ daily interactions across 3 hospitals) by building context-aware **multi-agent prompt pipeline**. Led 3-person ML team. [LangChain, GPT-4, AWS, Azure, FastAPI, LangSmith, DSPy]
- **LLM Evaluation and Safety Pipeline:** Built **RAGAS evaluation harness** measuring faithfulness, relevance, and context precision across 5,000+ sessions; implemented confidence-threshold HITL gating for low-certainty extractions; established eval-driven iteration loop for systematic prompt optimization via DSPy. [RAGAS, DSPy, LangSmith, HITL]
- **RLHF & Alignment Engineering (Qwen 2.5 72B):** Designed full alignment pipeline for medical LLM — fine-tuned **Qwen 2.5 72B** via LoRA/QLoRA for instruction following, then applied **DPO** on clinician-preference pairs to align outputs with diagnostic reasoning; trained **reward model** on domain-specific human preference data scoring safety, accuracy, and clinical relevance. Deployed on **Modal + RunPod GPU** infrastructure. [PyTorch, Unsloth, LLaMA Factory, Modal, RunPod, Qwen]
- **HIPAA-Compliant Infrastructure:** Delivered **99.99% uptime** for 500+ users across **Azure (Container Apps)** and **AWS (SageMaker)**; passed 2 regulatory audits with zero findings. [GitHub Actions, CI/CD, MLflow, Azure, AWS]
- **Clinical NLP Pipeline:** Achieved **\$150K annual savings** (85% manual review reduction) by building NLP pipeline with **97% accuracy** on 10K+ monthly documents. [spaCy, PyTorch, AWS SageMaker, Azure Cognitive Services]

Machine Learning Engineer

Jan 2023 - Aug 2023

Krut AI (Snive), Bengaluru

- **Generative Image Pipeline:** Reduced inference latency **8s to 1.2s (85% reduction)** (50K+ images/month) via **TensorRT**, Fast UNet, and **VAE optimizations** on **AWS G5 instances**. Mentored 2 junior engineers. [Stable Diffusion XL, TensorRT, ControlNet, AWS G5, ONNX, MLflow]
- **GPU Cost Optimization & MLOps Pipeline:** Reduced GPU costs **\$32K/month** via **mixed precision** and gradient checkpointing (SSIM >0.95 maintained); automated model versioning and deployment with **MLflow + GitHub Actions + ONNX** runtime optimization cutting deployment from **3 days to 4 hours**. [MLflow, GitHub Actions, ONNX, Docker, AWS]

Associate Data Scientist

Mar 2022 - Jan 2023

Lincode Labs Inc., Bengaluru

- **Event-Driven Architecture:** Achieved **\$2M/year infrastructure savings** (450M req/month across 15 global facilities) by replacing REST polling with **Server-Sent Events** on **AWS (EC2, S3, SQS)**. [Python, REST APIs, SSE, AWS]
- **CV/OCR Model Delivery & Monitoring:** Delivered **92% accuracy** models for 8 international clients (**0.24s inference**) via ONNX runtime optimization; monitored across **15 global facilities** via Prometheus + SSE streaming at **450M req/month**; reduced support tickets **60%** by training 20+ client engineers. [OpenCV, OCR, PyTorch, YOLO, ONNX, TensorFlow, Prometheus, SSE]

TECHNICAL SKILLS

GenAI and LLMs	GPT-4, Claude 3.5, Gemini Pro, LLaMA 3, Qwen 2.5, RAG, GraphRAG, Multi-Agent Systems, Knowledge Distillation, Prompt Engineering, Fine-tuning (LoRA/QLoRA), RLHF, DPO, PPO, Reward Modeling, Alignment Engineering, LLM-as-Judge, Confidence-Gated Evaluation, Agentic Workflows, Reasoning Models, Agent Harness Design
Orchestration and Protocols	LangChain, LangGraph, Llamaindex, Google ADK, Semantic Kernel, DSPy, NeMo Guardrails, RAGAS, MCP (Model Context Protocol), Provenance Tracking, Reranking, Agent Harness Design, Confidence-Gated HITL
LLM Inference and LLMOps	vLLM, TensorRT-LLM, SGLang, LMDeploy, Ollama, Unsloth, LLaMA Factory, High-throughput Batching, Edge Deployment, LangSmith, Prometheus, MLflow, CI/CD, MinerU VLM, Multi-provider LLM Abstraction, GitHub Actions
Cloud Platforms	AWS (SageMaker, ECS, Lambda, Bedrock, SQS, S3, G5), Azure (Container Apps, Cognitive Services, Key Vault), GCP (Vertex AI, Cloud Run, Gemini), Modal, RunPod
Vector and Data Stores	Neo4j, Qdrant, Pinecone, PostgreSQL, MongoDB, Redis, Elasticsearch, MiniO
ML / CV / Speech	PyTorch, TensorFlow, BERT, Stable Diffusion XL, YOLO, Whisper, ONNX, Quantization, spaCy, Microsoft Presidio
Infrastructure and Tools	Docker, Kubernetes, Terraform, GitHub Actions, FastAPI, REST, gRPC, Alembic, aiosqlite, SQLAlchemy 2, Prometheus
Programming Languages	Python, TypeScript, Rust, Go, JavaScript, HTML, CSS

EDUCATION

M.Sc. Big Data Analytics	St. Joseph's University, Bengaluru	2020 - 2022
B.C.A. Computer Applications	Seshadripuram College, Bengaluru	2016 - 2019

PROJECTS AND RECOGNITION

Distributed Inference Service

github.com/rakshit176/distributed-inference

Multi-region ML serving at 5K req/sec with 99.9% uptime across AWS and GCP. [Ray Serve, Redis, Prometheus, Terraform]

GSC Enterprise AI Search

Releasing publicly

1M+ graph nodes, 50M embeddings, hybrid graph + vector retrieval at p95 <300ms. [Neo4j, Qdrant, AWS Bedrock, LangSmith]

IDP Supply Chain Agent (Scania)

Client work, not yet public

5-stage confidence-gated agent harness for shipping document intelligence; p95 <300ms hybrid NL-to-SQL + semantic search; **73% HITL reduction** with zero critical-field errors. [AWS Bedrock, Claude, Elasticsearch, MinerU VLM, FastAPI, Docker]

Awards: Winner - Gruve AI Hackathon (IT Help Desk) | Winner - August AI Hackathon (AI Doctor Analyzer)

Publication: Co-author, IEEE ACAAI 2022 - "Water Quality Prediction using BPNN, SVR, and LSTM"

MCP Foundation Development Contributor - Open-source Model Context Protocol ecosystem